Statistical Significance in Model Comparison

Is a measured improvement real—or sampling noise?

Model Evaluation · Foundations of Machine Learning

Performance Estimates Are Random Variables

A test score is a statistic — It depends on which observations landed in the test set.

Resample, remeasure — A different split gives a different number for the same procedure.

Implication — "91% vs. 89%" is incomplete without a sense of uncertainty.

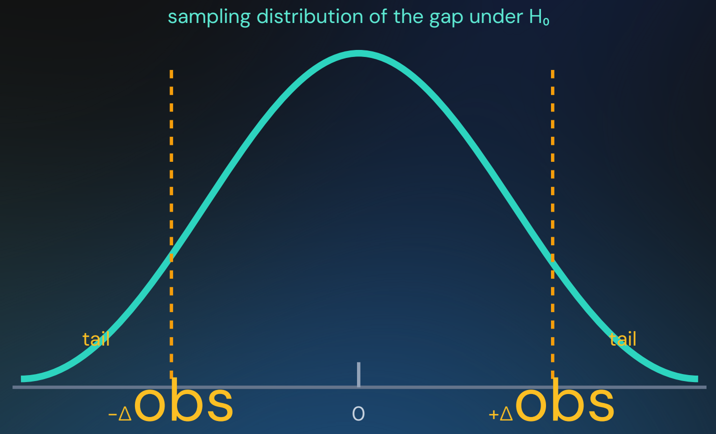
$$\widehat{\mathrm{Acc}} = \frac{1}{n}\sum_{i=1}^{n} \mathbf{1}[\hat{y}_i = y_i]$$

The Null-Hypothesis Frame

H_0 — The models have equal true performance; the observed gap Δ is noise.

p-value — Probability of a gap at least this large, if H_0 were exactly true.

Not what it means — It is not the probability that H_0 is true, and 0.05 is a convention, not a cliff.



McNemar's Test Uses Paired Disagreements

Same test examples — Each row has a prediction from model A and model B on the identical case.

Only discordant cells matter — n_{01} = A wrong / B right; n_{10} = A right / B wrong.

Ignore agreements — Both-right (n_{11}) and both-wrong (n_{00}) do not distinguish the models.

	B correct	B wrong
A correct	n_{11} both right	n_{10} A only
A wrong	n_{01} B only	n_{00} both wrong

$$\chi^2 = \frac{(|n_{01} - n_{10}| - 1)^2}{n_{01} + n_{10}}$$

Worked Example: McNemar's Test

	B correct	B wrong
A correct	400	8
A wrong	22	70

500 shared test examples. B corrects 22 cases A misses; A corrects only 8 cases B misses.

- $\chi^2 = \frac{(|22-8|-1)^2}{22+8} = \frac{169}{30} = 5.63$
- $p \approx 0.018$ (χ^2 with 1 df)
- Exact binomial on $n_{01} = 22$ of 30: $p \approx 0.016$
- Both < 0.05: reject H_0 — model B's edge is unlikely to be chance

Paired Tests Across CV Folds

Pair the folds — Both models are evaluated on the same held-out fold each time.

Test differences — Analyze d_i = score A_i − score B_i for each of k folds, not the raw scores.

Caveat — Overlapping training folds violate full independence and can make ordinary t-tests optimistic.

$$t = \frac{\bar{d}}{s_d/\sqrt{k}}$$

$\bar{d}$ = mean of the k paired differences. s_d = their sample standard deviation. Large |t| ⇒ the mean gap is large relative to how much it varies across folds.

Statistical vs. Practical Significance

Statistically significant

The observed effect is unlikely under the null model — a claim about surprise, not size.

Practically meaningful

The effect is large enough to justify added complexity, cost, latency, or risk.

Report both

Give the effect size and an interval, not only a p-value.

Model Comparison Checklist

Pair observations

Exploit shared test examples (McNemar) or shared folds (paired t-test).

Quantify uncertainty

Report a p-value or confidence interval, not just two point estimates.

Control multiplicity

Do not cherry-pick one winner from many tested models or metrics.

Judge value

Weigh the effect size against deployment cost and risk.

Trustworthy Comparisons

Scores vary

One number is one sample from a distribution.

Pair wisely

Use shared examples or folds; McNemar or paired t-test.

Size matters

Practical significance ≠ statistical significance.

Next: common evaluation failures — leakage and class imbalance.